

Read all of the questions carefully, and answer fully. Show work whenever possible.

Question 1. Data on the relationship between smoking and lung cancer are presented in the table to the right. You may leave your answers as fractions, but you may need to convert them to decimals to compare them.

	Lung cancer		Total
	No cancer	Cancer	
Smoking			
Non smoker	799	11	810
Heavy smoker	157	33	190
Total	956	44	1000

A. What proportion of people smoke heavily - that is, $p(\text{Heavy smoker})$? Is this a conditional or marginal probability?

B. What is the probability of developing lung cancer - that is, $p(\text{Cancer})$?

C. What is the probability that a person known to be a heavy smoker will get lung cancer - that is, $p(\text{Cancer} \mid \text{Heavy smoker})$

D. What is the probability of randomly selecting a person who is both a heavy smoker and has lung cancer - that is, $p(\text{Heavy smoker and Cancer})$?

E. Is lung cancer independent of smoking, or does lung cancer depend on smoking? Which probabilities did you compare to decide this?

F. If a patient is diagnosed with lung cancer, what is the probability that he or she is a heavy smoker - that is, $p(\text{Heavy smoker} \mid \text{Cancer})$?

2. Identify the variable type (numeric or non-numeric). For numeric variables, **also** indicate whether they are continuous or discrete. Give the measurement scale (interval, ratio, nominal, or ordinal) for each.

Variable	Type	Measurement scale
Number of eggs in a nest		
Calendar years		
Heights of shrubs		
Seasons		

3. A small team of researchers is designing a study of diabetes. They want to enroll a large number of diabetes patients in their study, but the cost of doing large numbers of blood tests in the hospital is prohibitive. They would like to use cheaper home test kits, but they are concerned that the kits might not be suitable for scientific research. They decide that to be useful for their study there can be no interobserver variation in using the kits, and none of the readings can be more than 0.5 mmol/L different from the true value. If either of these problems occurs, they will need to find a different instrument.

To test whether the test kits work well enough to use, one of the researchers (Bob) volunteers to be the blood source. They draw a single sample of blood from Bob, and the other three researchers (Laine, Andrea, and Kaitlin) test his blood sugar levels four times each with a home test kit. They send the remainder of the blood sample to the lab. The lab results show that Bob's blood sugar level is 5.0 mmol/L, which they use as the true blood sugar level.

The data from these tests are in the file "Question3.mtw" on the class web site.

A. Is there any biological variation in the data? Explain why or why not. What else besides biological variation could cause variability in the data?

B. Make an individual value plot with groups, using "Blood sugar" as the "Graph variable", and "Observer" as the "Categorical variable for grouping". Add a mean symbol (the same way you added bars for the geyser data in the second activity). Based on this graph, is there interobserver variation? Why or why not?

C. Which observer has the most biased measurements? Who has the least precise measurements?

D. Are there any observations that are more than 0.5 mmol/L from the true blood sugar level? Whose measurements are they?

E. Based on the data, is it possible to use this instrument well enough for the study? What could they do to improve their use of the home test kit so that it's acceptable for their study?

F. If you wanted to make a summary graph of their blood sugar data, would it be better to use an interval plot or a pie chart? Why?

4. Rachel wanted to determine whether two different generic brands of aspirin, Brand A and Brand B, were accurately representing the sizes of their pills. Both bottles were supposed to be 100 mg, so she weighed 100 pills of each brand on a balance. Her data are in the file "Question4.mtw" on the course web site.

A. Make histograms for BrandA and BrandB. What is the shape of each distribution (note: if there appear to be too many or too few bars to tell, you may change the number of bins. If you do change the number of bins, note how many bins you used in your answer)?

B. Calculate the mean, standard deviation, and standard error for each brand, and report them in the summary statistics table to the right.

Summary statistics

Statistic	BrandA	BrandB
Mean		
s		
$s_{\bar{x}}$		

Confidence intervals

Brand	Lower	Upper
BrandA		
BrandB		

C. Calculate the 95% confidence interval and report the lower and upper limits in the confidence intervals table on the previous page. You can use MINITAB for this. What degrees of freedom was used in these confidence intervals?

D. Are either of the manufacturers producing pills that are not 100 mg in size on average? How do you know?

E. Why do the confidence intervals differ in size for these two brands?

F. If you increased the sample size for the brand with the wider confidence interval to make it narrower, would you be more likely to contain the true mean within the interval? Why or why not?

G. Are you sure that using symmetrical confidence intervals is appropriate for these data? Why or why not?

H. When you take a medication, it's important that each dose is correct. Under-dosing can lead to ineffective treatment, and over-dosing can lead to toxicity. If you had to pick one of the brands to take, based on the data presented here which brand would be better?

5. In file "Question5.mtw" are examples of three variables we might encounter in the biological sciences: ages of trees in a forest (in years), lengths of axons from nerve cells in the brain (in mm), and gene expression levels (i.e. the amount of protein being produced by a gene) for 1000 genes on a chip.

A. Make a histogram of each, and describe their shapes.

B. Why would it be more common to have right-skewed variables than left-skewed variables in the biological sciences?

C. Would a box plot or interval plot be more appropriate as a summary graph for these variables? Why?